Smart Agriculture Data Analysis Project

1. Project Name

Smart Agriculture Crop Monitoring and Yield Optimization using Excel

2. Dataset Description

The dataset used is a Smart Agriculture dataset containing 200 records with mixed data types such as integer, float, date, and string. It includes crop, soil, weather, irrigation, pest, and yield information collected from multiple farms.

Dataset Screenshot:

Total Records: 200
Total Columns: 13

Farm_ID	Crop_Type	Region	Soil_Type	Sowing_Date	Harvest_Date	Soil_Moisture_Percent	Rainfall_mm	Temperature_Celsius	Fertilizer_Used_kg	Irrigation_Type	Pest_Incidence_Level	Crop_Yield_kg
1	Sugarcane	West	Sandy	2023-02-13 0:00:00	2023-07-10 0:00:00	38.59	164.78	17.83	39.73	Rainfed	Medium	1197.16
2	Rice	West	Clay	2023-01-29 0:00:00	2023-05-04 0:00:00	19.22	133.5	27.91	51.94	Canal	High	3279.69
3	Cotton	West	Clay	2023-01-12 0:00:00	2023-06-30 0:00:00	25.42	76.9	38.8		Sprinkler	Medium	2583.35
4	Maize	East	Sandy	2023-06-04 0:00:00	2023-09-17 0:00:00	57.13	185.82	28.93	13.83	Canal	Low	1447.84
5	Cotton	North	Black	2023-04-02 0:00:00	2023-08-19 0:00:00	19.51	155.94	24.19	84.3	Rainfed	Medium	3666.61
6	Cotton	West	Clay	2023-06-05 0:00:00	2023-11-06 0:00:00	54.62	83.27	22.8	68.52	Rainfed	Medium	590.74
7	Cotton	West	Loamy	2023-05-29 0:00:00	2023-09-30 0:00:00	19.84	152.04	22.53	57.81	Sprinkler	Low	4404.73
8	Rice	East	Loamy	2023-03-19 0:00:00	2023-08-29 0:00:00	36.05	74.14	27.65	22.36	Canal	Low	3649.02
9	Maize	South	Black	2023-03-03 0:00:00	2023-06-06 0:00:00	36.62	125.19	36.52	47.43	Canal	Medium	3069.76
10	Rice	East	Clay	2023-05-18 0:00:00	2023-09-25 0:00:00	49.28	128.59	27.58	97.59	Rainfed	Low	4783.6
11	Sugarcane	East	Black	2023-01-24 0:00:00	2023-05-28 0:00:00	53.67	117.64	39.53	6.45	Rainfed	High	608.54
12	Wheat	North	Black	2023-03-11 0:00:00	2023-08-16 0:00:00	27.43	31.53	30.25	45.67	Sprinkler	High	4399.48
13	Rice	East	Sandy	2023-03-21 0:00:00	2023-06-19 0:00:00	24.93	233.55	19.66	98.84	Drip	High	3765.48
14	Maize	South	Loamy	2023-02-25 0:00:00	2023-06-16 0:00:00	19.53	190.6	23.72	31.43	Rainfed	Low	4611.4
15	Cotton	South	Sandy	2023-03-17 0:00:00	2023-07-06 0:00:00	11.42	212.66	20.42	32.57	Drip	Low	1221.34
16	Maize	West	Clay	2023-06-19 0:00:00	2023-10-30 0:00:00	16.52	92.05	38.06	71.25	Sprinkler	Low	3426.85
17	Sugarcane	North	Sandy	2023-05-20 0:00:00	2023-08-22 0:00:00	59.89	123.03	16.16	37.37	Canal	High	1533.78
18	Cotton	West	Black	2023-01-29 0:00:00	2023-05-03 0:00:00	26.24	52.61	28.09	16.53	Rainfed	Low	1353.52
19	Rice	East	Black	2023-06-19 0:00:00	2023-11-07 0:00:00	42.42	26.63	17.64	60.75	Drip	High	4652.94
20	Rice	North	Clay	2023-03-27 0:00:00	2023-08-03 0:00:00	39.76	208.32	35.41	83	Canal	Low	2763.76
21	Maize	North	Black	2023-02-27 0:00:00	2023-07-02 0:00:00	42.64	197.25	16.91	89.47	Canal	High	1433.4
22	Wheat	North	Loamy	2023-06-06 0:00:00	2023-09-26 0:00:00	31.39	96.36	21.04	37.33	Sprinkler	Medium	3202.32
23	Rice	South	Clay	2023-02-02 0:00:00	2023-07-11 0:00:00		179.25	39.91	38.07	Sprinkler	Low	895.86
24	Rice	North	Clay	2023-01-27 0:00:00	2023-07-08 0:00:00	40.8	222.31	36.16	19.36	Sprinkler	High	4925.74
25	Cotton	East	Loamy	2023-03-17 0:00:00	2023-07-26 0:00:00	11.3	203.43	29.8	27.47	Drip	Low	2305.93
26	Wheat	East	Clay	2023-06-13 0:00:00	2023-09-29 0:00:00	16.44	164.6	16.12	57.95	Drip	Low	1684.85
27	Wheat	North	Loamy	2023-03-18 0:00:00	2023-07-15 0:00:00	39.63	135.05	25.78	73.63	Drip	High	2650.88

3. Objectives

O1. Analyze crop yield based on Soil & Weather conditions.
O2. Study the impact of soil moisture, rainfall, and temperature on crop yield.
O3.Analyze & compare agricultural productivity across different regions.

4. Tasks Based on Objectives

Objective 1 (O1): Analyze crop yield based on Soil & Weather conditions.

O1 – Task 1: Create Pivot Table to compare Crop Yield vs Soil Moisture.
O1 – Task 2: Analyze crop-wise yield distribution using Pivot Table.

Objective 2 (O2): Study the impact of soil moisture, rainfall, and temperature on crop yield

O2 – Task 1: Create Pivot Table to analyze average crop yield by soil moisture levels.
O2 – Task 2: Use Pivot Table to study rainfall and temperature impact on crop yield.
O2 – Task 3: Generate Pivot Charts to identify key environmental factors affecting yield.

Objective 3 (O3): Analyze & Compare agricultural productivity across different regions

O3 – Task 1: Create Pivot Table for region-wise crop yield analysis.
O3 – Task 2: Compare irrigation types using Pivot Table and Bar Chart.
O3 – Task 3: Evaluate soil type influence on yield performance.

5. Data Cleaning

Before performing analysis, data cleaning was conducted. Duplicate records were removed, missing values handled, date formats corrected, and data types validated.

6. Achieving the Objectives

Objectives O1 and O2 were achieved using Pivot Tables and Pivot Charts. Objectives O3 and O4 were achieved using Pivot Tables, charts, and dashboard tools.

7. Objective-wise Analysis

Objective	Graph Used	Pivot Table	Approach
O1	Combo Charts	Yes	Pivot-based crop yield analysis
O2	Column Charts	Yes	Soil & weather impact analysis
O3	Column/Bar Charts	Yes	Regional & irrigation comparison

8. Dashboard

The dashboard includes Crop Yield Summary, Soil & Weather Impact Overview, Region-wise and Irrigation-wise Yield Analysis, KPI Metrics, and interactive slicers.